

Sustainability A/B Testing Project: Paperless Billing Nudge

Does a “green nudge” message in the online banking portal actually increase paperless billing enrollment, and is it safe to ship?

Overview

This project simulates and analyzes an A/B test built around the type of experimentation work a bank's marketing analytics or growth team would run to support a customer sustainability initiative. It uses a fully synthetic dataset (no real institution's data) to demonstrate a complete, decision-ready experimentation workflow, not just a significance test, but the full set of checks a real launch decision requires: was the experiment set up correctly, did it work, was it safe, and who did it work best for.

The Business Question

A retail bank wants to increase paperless billing adoption as part of its sustainability strategy. One proposed lever: a green-themed nudge message shown to customers in their online banking portal, encouraging them to switch. Before rolling this out broadly, four questions need answers:

1. Does the nudge causally increase paperless enrollment, and by how much?
2. Is the effect big enough to be worth acting on, or could an observed lift just be noise from an underpowered test?
3. Does it work better for some customer segments than others (age, channel), informing a targeted vs. broad rollout?
4. Does it come at a cost, such as more unsubscribes or complaints, that would make it a bad trade even if adoption goes up?

Approach

- **Synthetic data simulation:** ~20,000 customers randomly assigned to control (standard messaging) or treatment (green nudge), with realistic demographic and channel variation and a built-in treatment effect (stronger for younger customers and push-notification users).
- **Power / sample-size analysis (a priori):** calculates the minimum sample size needed to reliably detect a 2.5-percentage-point lift at 80% power, checked against the actual sample size. This is the step that determines whether a null result means the nudge doesn't work, or the test just wasn't big enough to tell.
- **Sample Ratio Mismatch (SRM) check:** chi-square test confirming the 50/50 randomization held, plus covariate balance checks (segment, channel) across groups.
- **Primary analysis:** two-sample proportions z-test on paperless enrollment, plus a logistic regression controlling for segment, channel, and age.
- **Guardrail metrics:** z-tests on unsubscribe rate and complaint rate, to confirm the nudge isn't causing enough friction to make it a bad trade even if adoption improves.

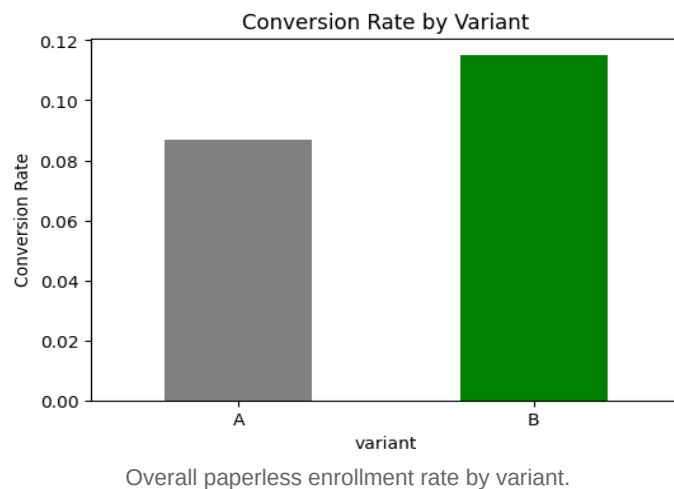
- **Subgroup analysis:** conversion broken out by customer segment and preferred channel.
- **Tech stack:** Python, pandas/NumPy, SciPy/statsmodels (proportions test, logistic regression, power analysis), matplotlib.

Results

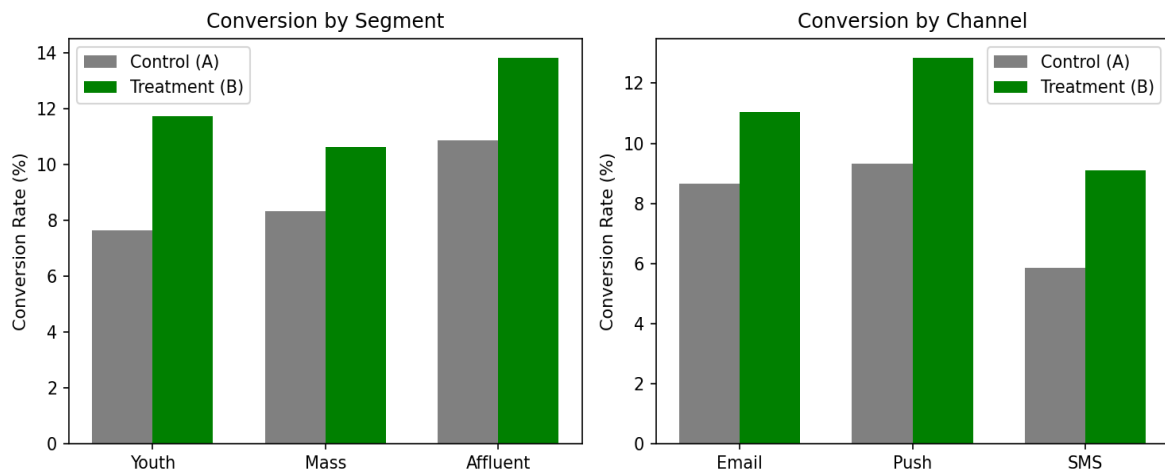
Check	Result
Power analysis	Required: 1,654/group vs. actual: 10,000/group → adequately powered (6x minimum)
SRM check	$p = 0.562$ → randomization held, segment/channel balance confirmed
Primary lift	Control: 8.71%, Treatment: 11.49%, uplift: +2.79 pp (+32% relative), $p < 0.001$
Logistic regression (adjusted)	Treatment odds ratio: 1.36 (+36% odds), $p < 0.001$
Guardrail: unsubscribe rate	A: 0.34%, B: 0.39%, $p = 0.291$ → pass
Guardrail: complaint rate	A: 0.18%, B: 0.18%, $p = 0.510$ → pass
Subgroup: segment	Youth +4.12 pp (+54% rel.) · Affluent +2.96 pp (+27%) · Mass +2.30 pp (+28%)
Subgroup: channel	Push +3.53 pp (+38%) · SMS +3.24 pp (+56% rel., small base) · Email +2.40 pp (+28%)

Age itself was not a significant predictor in the regression ($p = 0.947$) once segment is controlled for. The segment variable captures age-related response differences, not raw age.

Conversion Rate: Control vs. Treatment



Conversion by Segment and Channel



Subgroup breakdown of conversion rate by customer segment (left) and preferred communication channel (right).

Business Recommendation

Roll out the green nudge broadly. The nudge produced a statistically significant 2.79 pp lift in paperless adoption (a 32% relative increase, $p < 0.001$), the experiment was well-powered to detect this effect (6x the minimum required sample), and both guardrail metrics held steady, with no evidence the nudge increases unsubscribes or complaints. This is a clean “ship it” result: the effect is real, it's large enough to matter, and it doesn't come at a customer-experience cost.

Prioritize rollout sequencing by segment and channel. The effect is not uniform. Youth customers (+54% relative) and push-notification users (+38%) respond meaningfully more than the affluent/mass segments or email channel. If rollout needs to be phased, start with youth + push-notification customers for the fastest measurable impact, then expand to the broader base.

One caveat worth flagging before this goes further: the SMS channel shows the largest *relative* lift (+56%) but represents only ~5% of the customer base in this simulation, so that subgroup estimate is the least precise of the group, worth a dedicated, adequately powered follow-up test on SMS specifically before treating it as a confirmed high performer.

Why This Project Matters

Experimentation is the core toolkit for product/growth data science roles. A working test isn't just “run a t-test,” it's confirming the test was set up correctly (SRM), sized correctly (power analysis) before trusting the result, checking it's safe to ship (guardrails), and understanding who it works for (subgroups). This project walks through all four, using a business scenario grounded in a real-world retail banking sustainability use case.

Note: this project uses a fully synthetic dataset generated in-notebook for demonstration and portfolio purposes. No real institution's data was used at any stage.